# Exact Outer Multiset Dimension of the Three-by-Three Path Grid Strips

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Independent AI differential audit through v0.3: author-reported PASS.
Windows PowerShell 5.1 reproduction of v0.3: author-supplied PASS.
Novelty: INCONCLUSIVE. Human peer review: NOT PERFORMED.
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**Abstract**

We give a computer-assisted exact classification of the outer multiset dimension of $G_n = P_3 \,\square\, P_3 \,\square\, P_n$, for $n \geq 2$. The value is 6 for $n = 3$, 5 for $n \in \{2, 4, 5, 6, 7, 8\}$, and 4 for $n \geq 9$. The lower bound $\operatorname{odim}(G_n) \geq 4$ uses the separately published, computer-assisted universal three-dimensional lower bound in the author's No. 59 preprint; it is not independently reproved here. The small exceptional cases are established by exhaustive finite certificates, while the infinite upper bound follows from four explicit landmarks, distance-sum identities, and two bounded integer collision checks. The result is neither formally verified nor human peer reviewed; novelty remains unconfirmed.

## 1 Definition, scope, and relation to earlier work

For a connected graph $G$ and $S \subseteq V(G)$, write $\mathcal{M}_S(v) = \{\{d_G(v, s) : s \in S\}\}$ for the distance *multiset*, with multiplicities retained and order forgotten. The set $S$ is *outer multiset resolving* if $\mathcal{M}_S(u) \neq \mathcal{M}_S(v)$ for all distinct $u, v \in V(G) \setminus S$. Its minimum cardinality is $\operatorname{odim}(G)$. The property is not generally monotone under adding landmarks; in particular, absence of a $k$-landmark solution alone does not establish absence of every smaller solution.

The two-path value $\operatorname{odim}(P_s \,\square\, P_t) = 3$ for $s \geq t \geq 2$ is due to Klavžar–Kuziak–Yero [1]. Farhan–Klavžar–Kuziak–Yero list extension to more than two path factors as Problem 10 in their 2026 survey [2]. This manuscript addresses only the subfamily $P_3 \,\square\, P_3 \,\square\, P_n$, not general three-path products.

The author's No. 58 [3] examines $P_2 \,\square\, P_3 \,\square\, P_n$. Our $G_2 \cong P_2 \,\square\, P_3 \,\square\, P_3$ is the *same graph* as the $n = 3$ finite instance in the author's report on No. 58. The value 5 is also recorded in the short-grid table of No. 59 [4]. The published theorem of No. 58 covers $n \geq 6$. We include $G_2$ as an overlap for completeness. No. 59 supplies the universal lower bound

$$\operatorname{odim}(P_a \,\square\, P_b \,\square\, P_c) \geq 4 \qquad (a, b, c \geq 2). \tag{1}$$

No. 59 is an author preprint with computer-assisted verification, not an independently formalized theorem of the external literature. Our reliance on it is explicit.

Identify $V(G_n)$ with $\{0, 1, 2\}^2 \times \{0, \ldots, n-1\}$. The graph distance is $d((x, y, z), (u, v, w)) = |x - u| + |y - v| + |z - w|$.

**Theorem 1.** *For every integer $n \geq 2$,*

$$\operatorname{odim}(P_3 \,\square\, P_3 \,\square\, P_n) = \begin{cases} 6, & n = 3, \\ 5, & n = 2 \text{ or } 4 \leq n \leq 8, \\ 4, & n \geq 9. \end{cases}$$

*The proof depends on the lower-bound result (1), finite exhaustive computation, and the exact finite certificates detailed below.*

## 2 Small parameters and finite certificates

For each $n = 2, \dots, 8$, the frozen C++17 verifier exhaustively tests all $\binom{9n}{4}$ four-landmark sets and finds none outer multiset resolving. For $n = 3$, it also tests all $\binom{27}{5}$ five-landmark sets, again finding none. This verifier represents a $k$-distance multiset by its first $k$ integer power sums. Newton's identities imply that these determine the multiset exactly; no probabilistic hash collision is accepted as equivalence because unordered-set equality compares the full power-sum array (the hash is only an index).

| $n$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| $\binom{9n}{4}$ | 3060 | 17550 | 58905 | 148995 | 316251 | 595665 | 1028790 |

The additional $n = 3$ five-set total is $\binom{27}{5} = 80730$. Explicit upper witnesses are recorded verbatim in `frozen/verify_finite.cpp`, for sizes 5 at $n = 2, 4, \dots, 8$ and size 6 at $n = 3$. Their distance multisets are checked directly. With (1) excluding every size at most three, these computations settle the minimum values for $n \le 8$. The source and expected outputs are frozen and accompanied by SHA256 digests.

## 3 Four-landmark constructions for all $n \ge 9$

For odd $n \ge 9$ let $c = 3$; for even $n \ge 12$ let $c = 6$. Set

$$S_n = \{(0,0,0), (0,0,1), (0,0,n-1), (0,2,c)\}. \tag{2}$$

At $n = 10$ use instead the exceptional witness

$$S_{10} = \{(0,0,0), (0,0,6), (0,2,8), (2,2,5)\}. \tag{3}$$

For $9 \le n \le 31$, the frozen Python certificate directly compares all sorted four-distance signatures outside $S_n$ and finds them distinct. For $n \ge 32$, the following lemmas reduce infinitely many possible collisions to small, explicit integer checks.

Let $p = x + y$, $a = 2x + y$, and write the distances to the four landmarks of (2) in the listed order as $A, B, C, D$. Then

$$(A, B, C, D) = (p + z,\ p + |z - 1|,\ p + n - 1 - z,\ x + 2 - y + |z - c|). \tag{4}$$

Their sum $T = A + B + C + D$ satisfies the exact identity

$$T = \begin{cases} n + c + 2 + 2a, & z = 0, \\ n + c + 2a, & 1 \le z < c, \\ n - c + 2(z + a), & z \ge c. \end{cases} \tag{5}$$

Here $0 \le a \le 6$.

**Lemma 1** (Near-prefix reduction)**.** *Suppose $n \ge 32$ and two distinct vertices outside $S_n$ have equal four-distance multisets. If either vertex has $z < c$, both have $z \le c + 7$. On this prefix, $C$ is strictly larger than each of $A, B, D$. Consequently any collision in this case is detected by equality of*

$$\big(p - z,\ \mathrm{sort}\big(A, B, D\big)\big)\,.$$

*The exact finite prefix check finds no such collision.*

*Proof.* By (5), a vertex with $z < c$ has $T \le n + c + 14$. For a vertex at $z' \ge c$, equality of the sums implies $n - c + 2z' \le n + c + 14$, whence $z' \le c + 7$. If both vertices have $z < c$, the same bound holds trivially. On $0 \le z \le c + 7 \le 13$ and $n \ge 32$ we have $C - A = n - 1 - 2z > 0$, and $C > B$ by inspection of $z = 0$ or $z \ge 1$. Moreover $C - D = n - 3 + 2y - z - |z - c| > 0$: for $z \le c$, it is at least $n - 3 - c$, and for $z \ge c$, at least $n - c - 17$. Thus $C$ is the unique maximum. Equal multisets must have $C = C'$, so $p - z = p' - z'$; the other three distances must agree as multisets. The frozen exact checker enumerates every vertex in this prefix except $(0,0,0), (0,0,1), (0,2,c)$ (the fourth landmark $(0,0,n-1)$ is outside the prefix). It checks 96 vertices for $c = 3$, and 123 for $c = 6$, and finds all indicated signatures unique. □

**Lemma 2** (Tail collision certificate)**.** *For $n \ge 32$, no two distinct vertices with $z, z' \ge c$ have equal four-distance multisets with respect to (2).*

*Proof.* Set $dx = x - x'$, $dy = y - y'$, and $t = p + z$. By the third case of (5), equality of sums implies

$$z' - z = 2dx + dy. \tag{6}$$

Subtract the common integer $t$ from each element of the two multisets. With $h = n - 1 - 2z$ this yields the necessary equality

$$\{\{0, -1, 2 - 2y - c, h\}\} = \{\{dx, dx - 1, dx + 2 - 2y' - c, h - 3dx - 2dy\}\}. \tag{7}$$

Since $dx, dy \in [-2, 2]$, the first three elements of the left multiset lie in $[-8, 0]$ and those on the right in $[-10, 2]$. Also $|3dx + 2dy| \leq 10$. If $|h| > 30$, the $h$-term is the unique extreme (largest if $h > 30$, smallest if $h < -30$), on both sides, forcing $h = h - 3dx - 2dy$. The sole solution of $3dx + 2dy = 0$ with $dx, dy \in [-2, 2]$ is $dx = dy = 0$; then (6) forces $z = z'$, contradicting distinctness.

It remains to consider $|h| \leq 30$. Since $h \equiv n - 1 \pmod 2$, there are 31 allowed even $h$ values when $n$ is odd ($c = 3$), and 30 allowed odd $h$ values when $n$ is even ($c = 6$). For each $c$, exhaust the 72 ordered *distinct* pairs $(x, y) \neq (x', y')$ in $\{0, 1, 2\}^2$, together with every such $h$. Sorting the four entries on both sides of (7) produces no equality among $72 \cdot 31 = 2232$ checks for $c = 3$, respectively $72 \cdot 30 = 2160$ checks for $c = 6$. This is a safe overapproximation: the checker does *not* require the corresponding $z, z'$ to be feasible grid coordinates. No floating-point arithmetic or random sampling is used. □

*Proof of Theorem 1.* Equation (1) excludes all outer multiset resolving sets of size at most three. The finite exhaustive computation and the explicit upper witnesses settle $2 \leq n \leq 8$. For $9 \leq n \leq 31$, the direct four-point certificate supplies an upper bound and (1) a matching lower bound. For $n \geq 32$, every pair of distinct exterior vertices is either covered by Lemma 1 or has both $z, z' \geq c$ and is covered by Lemma 2. Thus (2) gives the upper bound four, which is sharp by (1). □

## 4 Reproducibility and limitations

The original exploration sources `verify_finite.cpp` (C++17) and `verify_infinite.py` (Python standard library), together with their two expected stdout records, are preserved byte for byte from v0.3 in the accompanying v1.0 reproducibility package. The first exhaustively enumerates the finite lower-bound cases, using exact integer power sums. The second checks finite instances and the two analytical collision reductions. The runner verifies the file inventory and SHA256 digests, compiles in a temporary directory, and compares the outputs with the frozen records. Release preparation reproduced both outputs on Linux.

The author's supplied reports record a scope-limited independent AI differential audit PASS through v0.3. The reported scope covers the model, finite and infinite claims, editorial changes, package integrity, and privacy. Earlier reports describe independent BFS checks for finite cases and a rederivation of the infinite collision reduction. The v0.3 reviewer's independent lower-bound tests were finite. The all-$n$ lower bound in this paper relies on No. 59. The raw reviewer materials are private.

The author's supplied transcript records a Windows PowerShell 5.1 reproduction PASS for the exact v0.3 package on 10 November 2026. It covers the inventory and SHA256 checks, the four frozen files, and both computational certificates. The reproducibility package includes a sanitized summary of that execution. The Windows result applies to v0.3. We checked the v1.0 release package on Linux.

The literature search is incomplete and the novelty status is INCONCLUSIVE. Human peer review and full formal verification have not been performed.

## References


[1] S. Klavžar, D. Kuziak, I. G. Yero, "Further Contributions on the Outer Multiset Dimension of Graphs," *Results in Mathematics* **78** (2023), Article 50. https://doi.org/10.1007/s00025-022-01829-8.

[2] M. Farhan, S. Klavžar, D. Kuziak, I. G. Yero, "Multiset resolvability parameters in graphs: A survey with new results and open problems," arXiv:2607.10311v1 (2026), Problem 10. https://arxiv.org/abs/2607.10311.

[3] R. Yonezu, "Exact Outer Multiset Dimension of the Three-Dimensional Grid Family $P_2 \Box P_3 \Box P_n$," author preprint No. 58, v1.0 (2026). GitHub release v1.0.

[4] R. Yonezu, "Outer Multiset Dimension of Three-Dimensional Grids: A Universal Lower Bound and an Exact Two-Parameter Classification," author preprint No. 59, v1.0 (2026), universal lower-bound theorem. GitHub release v1.0.